Exercise 2

EMPLOYEES

Id	First_name	Last_name	department	salary	hire date	City
1	John	Doe	IT	55 000	2018-08-15	New York
2	Jane	Smith	HR	48 000	2019-07-19	Chicago
3	Mike	Johnson	Finance	60 000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	63 000	2021-03-25	New York
5	David	White	Marketing	52 000	2016-04-10	San Francisco
6	Emily	Davies	IT	62 000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59 000	2019-10-01	Houston
8	Jessica	Moore	HR	51 000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53 900	2022-06-01	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	San Francisco

1. Count() Function
Write a SQL query to find the total number of employees in the company

```
SELECT COUNT ID AS number_of_employees
FROM employee;
```

number_of_employee
10

2. SUM () Function
Write a SQL query to find the total salary paid in IT department

```
SELECT SUM(salary) AS total_it_salary, department
FROM employees
WHERE department = 'IT';
```

total_it_salary	department
220 000	IT

3. AVG Function
Write a SQL query to calculate the average salary of employees in HR department

```
SELECT AVG(Salary) AS Average_HR_Salary, department
FROM employees
WHERE department = 'HR'
```

Average_HR_Salary	department
49 500	HR

4. MIN() and MAX Functions

Write a SQL query to find the highest & Lowest salary in the company

```
SELECT MIN(Salary) AS lowest_salary
FROM employees;
```

→

Lowest_salary
48 000

```
SELECT MAX(Salary) AS highest_salary
FROM employees;
```

Highest_salary
62 000

5. Group By Statement

Write a SQL query to group employees by department & display the total salary paid in each department

```
SELECT department,
       SUM(Salary) AS total_salary
FROM employees
GROUP BY department;
```

→

Department	total_salary
IT	220 000
HR	99 000
Finance	119 000
Marketing	105 000

6. GROUP BY and COUNT

Write a SQL query to count how many employees work in each city

```
SELECT Count(Id) AS employees_per_city
       City
FROM employees
Group BY City;
```

City	employees_per_city
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

7. Group By and Order By

Write a SQL query to employees by department, calculate the average salary in each department, and order the results in descending order of AVG salary

```
SELECT Avg(salary) AS Avg salary_depmt, department
FROM employees
Group by department
ORDER By Avg_sal_dep DESC,
```

Avg_Salary_dep'	department
59500	Finance
55 000	IT
52590	Marketing
49500	HR

8. Having Clause
Write a SQL query to find department where the total salary paid exceed 100 000

```
SELECT    sum(salary) AS total-salary, department
FROM      employees
GROUP BY  department
HAVING    sum(salary) > 100 000;
```

Department	total-salary
IT	220 000
Finance	119 000
Marketing	105 000

9. Combining GROUP BY, HAVING and ORDER BY
Write a SQL query to list cities where more than one employees works, ordered by the number of employees in descending order

```
SELECT    city, COUNT(ID) AS employee_city
FROM      employees
GROUP BY  city
HAVING    COUN(ID) >1
ORDER BY  employee_city DESC;
```

City	employee_city
Chicago	3
New York	2
Los Angels	2
San Franisce	2

10. Combining Aggregate Functions
Write a SQL query to find the department with the highest average salary.

```
SELECT    AVG(Salary) AS High_Salary, department
FROM      employees
Group BY  department
HAVING    Avg(Salary) MAX
ORDE BY   High-salary DESC
Limit     1;
```